
CHAPTER 1

Fundamentals of Statistics

1.1 INTRODUCTION

In the modern world of information and communication technology, the importance of statistics is very well recognized by all the disciplines. Statistics has originated as a science of statehood and found applications slowly and steadily in Agriculture, Economics, Commerce, Biology, Medicine, Industry, planning, education and so on. As of today, there is no other human walk of life, where statistics cannot be applied.

Statistics is concerned with the scientific method of collecting, organizing, summarizing, presenting and analyzing statistical information (data) as well as drawing valid conclusion on the basis of such analysis. It could be simply defined as the "science of data". Thus, statistics uses facts or numerical data, assembled, classified and tabulated so as to present significant information about a given subject. Statistic is a science of understanding data and making decisions in the face of randomness.

The study of statistics is therefore essential for sound reasoning, precise judgment and objective decision in the face of up- to- date accurate and reliable data. Thus many researchers, educationalists, business men and government agencies at the national, state or local levels rely on data to answer operations and programs.

In this chapter, you begin by learning five basic words—population, sample, variable, parameter, and statistic (singular)—that identify the fundamental concepts of statistics. These five words, and the other concepts introduced in this chapter, help you explore and explain the statistical methods discussed in later chapters.

1.2 The First Three Words of Statistics

You’ve already learned that statistics is about analyzing things. Although *numbers* was the word used to represent things in the opening of this chapter, the first three words of statistics, *population*, *sample*, and *variable*, help you to better identify what you analyze with statistics.

Population

CONCEPT. All the members of a group about which you want to draw a conclusion.

EXAMPLES. All DZ citizens who are currently registered to vote, all patients treated at a particular hospital last year, the entire daily output of a cereal factory’s production line.

Sample

CONCEPT. The part of the population selected for analysis.

EXAMPLES. The registered voters selected to participate in a recent survey concerning their intention to vote in the next election, the patients selected to fill out a patient satisfaction questionnaire, 100 boxes of cereal selected from a factory's production line.

Variable

CONCEPT. A characteristic of an item or an individual that will be analyzed using statistics.

EXAMPLES. Gender, the party affiliation of a registered voter, the household income of the citizens who live in a specific geographical area, the publishing category (hardcover, trade paperback, mass-market paperback, textbook) of a book, the number of televisions in a household.

INTERPRETATION. All the variables taken together form the data of an analysis. Although people often say that they are analyzing their data, they are, more precisely, analyzing their variables. (Consistent to everyday usage, the authors use these terms interchangeably throughout books.)

You should distinguish between a variable, such as gender, and its value for an individual, such as male. An observation is all the values for an individual item in the sample. For example, a survey might contain two variables, gender and age. The first observation might be male, 40. The second observation might be female, 45. The third observation might be female, 55. A variable is sometimes known as a column of data because of the convention of entering each observation as a unique row in a table of data. (Likewise, some people refer to an observation as a row of data.)

Variables can be divided into the following types:

	Categorical Variables	Numerical Variables
Concept	The values of these variables are selected from an established list of categories.	The values of these variables involve a counted or measured value.
Subtypes	<i>Nominal scale</i> is a naming scale, where variables are simply "named" or labeled, with no specific order. <i>Ordinal scale</i> has all its variables in a specific order, beyond just naming them.	<i>Discrete.</i> values are counts of things. <i>Continuous.</i> values are measures and any value can theoretically occur, limited only by the precision of the measuring process.
Examples	Gender, a variable that has the categories "male" and "female." Academic major, a variable that might have the categories "English," "Math," "Science," and "History," among others.	The number of people living in a household, a discrete numerical variable. The time it takes for someone to commute to work, a continuous variable.

1.3 The Fourth and Fifth Words

After you know what you are analyzing, or, using the words of the previous Section, after you have identified the variables from the population or sample under study, you can define the parameters and statistics that your analysis will determine.

Parameter

CONCEPT. A numerical measure that describes a variable (characteristic) of a population.

EXAMPLES. The percentage of all registered voters who intend to vote in the next election, the percentage of all patients who are very satisfied with the care they received, the mean weight of all the cereal boxes produced at a factory on a particular day.

Statistic

CONCEPT. A numerical measure that describes a variable (characteristic) of a sample (part of a population).

EXAMPLES. The percentage of registered voters in a sample who intend to vote in the next election, the percentage of patients in a sample who are very satisfied with the care they received, the mean weight of a sample of cereal boxes produced at a factory on a particular day.

INTERPRETATION. Calculating statistics for a sample is the most common activity because collecting population data is impractical in most actual decision-making situations.

1.4 The Branches of Statistics

You can use parameters and statistics either to describe your variables or to reach conclusions about your data. These two uses define the two branches of statistics: **descriptive statistics** and **inferential statistics**.

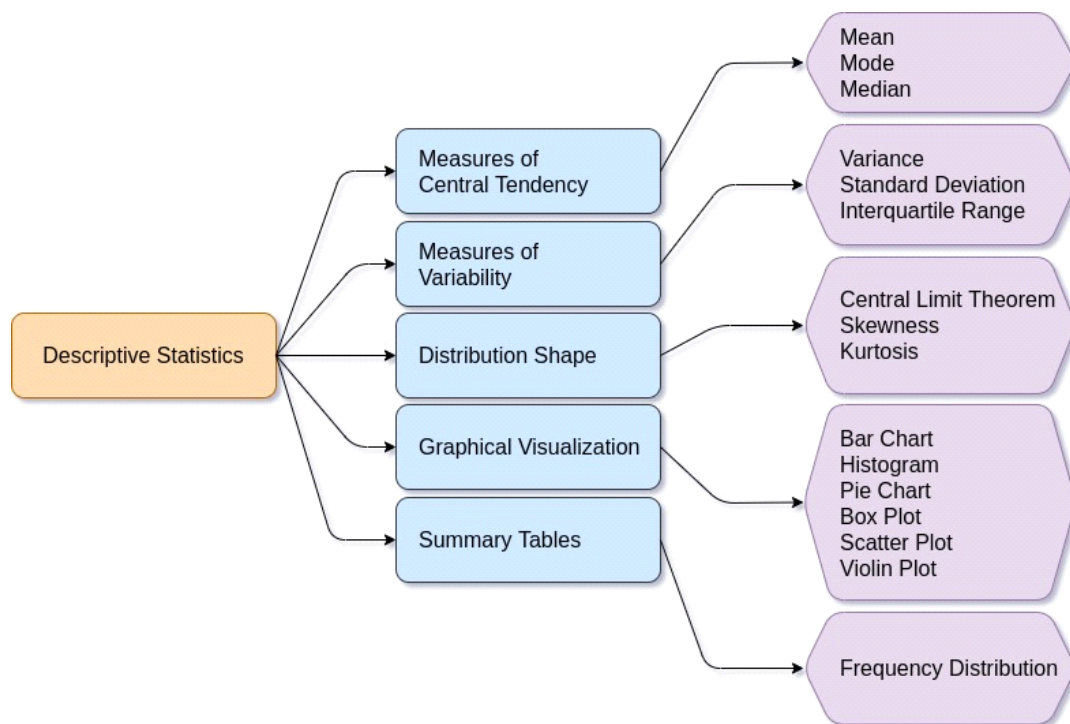
1.4.1 Descriptive Statistics

CONCEPT. The branch of statistics that focuses on collecting, summarizing, and presenting a set of data.

EXAMPLES. The mean age of citizens who live in a certain geographical area, the

mean length of all books about statistics, the variation in the weight of 100 boxes of cereal selected from a factory's production line.

INTERPRETATION. You are most likely to be familiar with this branch of statistics because many examples arise in everyday life. Descriptive statistics serves as the basis for analysis and discussion in fields as diverse as securities trading, the social sciences, government, the health sciences, and professional sports. Descriptive methods can seem deceptively easy to apply because they are often easily accessible in calculating and computing devices. However, this easiness does not mean that descriptive methods are without their pitfalls, as Chapters of Presenting Data in Charts and Tables and Descriptive Statistics explain.

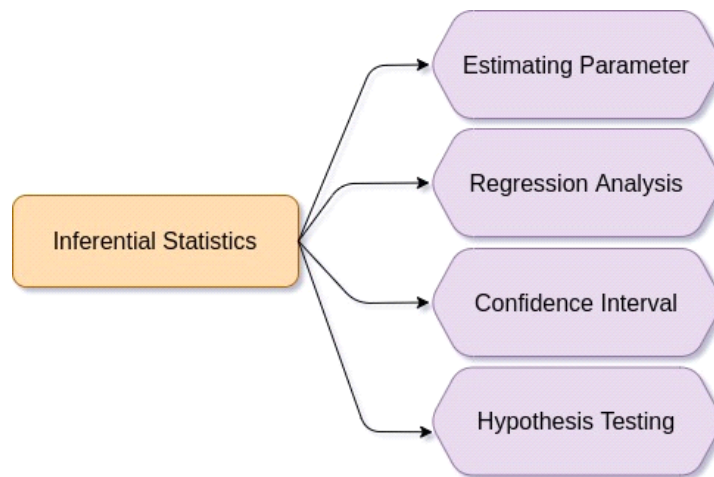


1.4.2 Inferential Statistics

CONCEPT. The branch of statistics that analyzes sample data to reach conclusions about a population.

EXAMPLE. A survey that sampled 1,264 women found that 45% of those polled considered friends or family as their most trusted shopping advisers and only 7% considered advertising as their most trusted shopping adviser. By using methods discussed in next Section, you can use these statistics to draw conclusions about the population of all women.

INTERPRETATION. When you use inferential statistics, you start with a hypothesis and look to see whether the data are consistent with that hypothesis. This deeper level of analysis means that inferential statistical methods can be easily misapplied or misconstrued, and that many inferential methods require a calculating or computing device.



1.4.3 USES OF STATISTICS

Statistics can be used among others for:

- 1) Planning and decision making by individuals, states, business organizations, research institution etc.
- 2) Forecasting and prediction for the future based on a good model provided that its basic assumptions are not violated.
- 3) Project implementation and control. This is especially useful in on-going projects such as network analysis, construction of roads and bridges and implementation of government programs and policies.
- 4) The assessment of the reliability and validity of measurements and general points significance tests including power and sample size determination.

1.5 STATISTICAL DATA

Data can be described as a mass of unprocessed information obtained from measurement of counting of a characteristics or phenomenon. They are raw facts that have to be processed in numerical form they are called **quantitative data**. For instance the collection of ages of students in a particular session is an example of this data. But when data are not presented in numerical form, they are called **qualitative data**. E.g.: status, sex, religion, etc.

Définition 1.5.1 *Statistical data are data obtained through objective measurement or enumeration of characteristics using the state of the art equipment that is precise and unbiased. Such data when subjected to statistical analysis produce results with high precision.*

1.5.1 SOURCES OF STATISTICAL DATA

1. *Primary data:* These are data generated by first hand or data obtained directly from respondents by personal interview, questionnaire, measurements or observation.

Statistical data can be obtained from:

- (i) Census – complete enumeration of all the unit of the population
- (ii) Surveys – the study of representative part of a population
- (iii) Experimentation – observation from experiment carried out in laboratories and research center.
- (iv) Administrative process e.g. Record of births and deaths.

ADVANTAGES

- Comprises of actual data needed
- It is more reliable with clarity
- Comprises a more detail information

DISADVANTAGES

- Cost of data collection is high
- Time consuming
- There may larger range of non response

2. *Secondary data:* These are data obtained from publication, newspapers, and annual reports. They are usually summarized data used for purpose other than the intended one. These could be obtain from the following:

- (i) Publication e.g. extract from publications

(ii) Research/Media organization

(iii) Educational institutions

ADVANTAGES

- The outcome is timely
- The information gathered more quickly
- It is less expensive to gather.

DISADVANTAGES

- Most time information are suppressed when working with secondary data
- The information may not be reliable

1.5.2 METHODS OF COLLECTION OF DATA

There are various methods we can use to collect data. The method used depends on the problem and type of data to be collected. Some of these methods include:

1. Direct observation
2. Interviewing
3. Questionnaire
4. Abstraction from published statistics.

DIRECT OBSERVATION

Observational methods are used mostly in scientific enquiry where data are observed directly from controlled experiment. It is used more in the natural sciences through laboratory works than in social sciences. But this is very useful studying small communities and institutions.

INTERVIEWING

In this method, the person collecting the data is called the interviewer goes to ask the person (interviewed) direct questions. The interviewer has to go to the interviewed personally to collect the information required verbally. This makes it different from the next method called questionnaire method.

QUESTIONNAIRE

A set of questions or statement is assembled to get information on a variable (or a set of variable). The entire package of questions or statement is called a questionnaire. Human beings usually are required to respond to the questions or statements on the questionnaire. Copies of the questionnaire can be administered personally by its user or sent to people by post. Both interviewing and questionnaire methods are used in the social sciences where human population is mostly involved.

ABSTRACTIONS FROM THE PUBLISHED STATISTICS

These are pieces of data (information) found in published materials such as figures related to population or accident figures. This method of collecting data could be useful as preliminary to other methods.

Other methods includes: Telephone method, Document/Report method, Mail or Postal questionnaire, On-line interview method, etc.

1.6 PRESENTATION OF DATA

When raw data are collected, they are organized numerically by distributing them into classes or categories in order to determine the number of individuals belonging to each

class. Most cases, it is necessary to present data in tables, charts and diagrams in order to have a clear understanding of the data, and to illustrate the relationship existing between the variables being examined.

1.6.1 FREQUENCY TABLE

This is a tabular arrangement of data into various classes together with their corresponding frequencies.

Procedure for forming frequency distribution

Given a set of observation $x_1, x_2, \dots, x_n$, for a single variable.

1. Determine the range $(R) = L - S$ where L = largest observation in the raw data; and S = smallest observation in the raw data.
2. Determine the appropriate number of classes or groups (K). The choice of K is arbitrary but as a general rule, it should be a number (integer) between 5 and 20 depending on the size of the data given. There are several suggested guide lines aimed at helping one decided on how many class intervals to employ.

Two of such methods are:

- (a) $K = 1 + 3.322 \times \log_{10}(n)$
 - (b) $K = \sqrt{n}$ where n = number of observations.
3. Determine the width (w) of the class interval. It is determined as $w = \frac{R}{K}$
 4. Determine the numbers of observations falling into each class interval i.e. find the class frequencies.

NOTE: With advent of computers, all these steps can be accomplishes easily.

SOME BASIC DEFINITIONS

Variable: This is a characteristic of a population which can take different values.

Basically, we have two types, namely: continuous variable and discrete variable. A continuous variable is a variable which may take all values within a given range. Its values are obtained by measurements e.g. height, volume, time, exam score etc. A discrete variable is one whose value change by steps. Its value may be obtained by counting. It normally takes integer values e.g. number of cars, number of chairs.

Class interval: This is a sub-division of the total range of values which a (continuous) variable may take. It is a symbol defining a class E.g. 0-9, 10-19 etc. there are three types of class interval, namely: Exclusive, inclusive and open-end classes method.

Exclusive method:

When the class intervals are so fixed that the upper limit of one class is the lower limit of the next class; it is known as the exclusive method of classification. E.g. Let some expenditures of some families be as follows: 0 – 1000, 1000 – 2000, etc. It is clear that the exclusive method ensures continuity of data as much as the upper limit of one class is the lower limit of the next class. In the above example, there are so families whose expenditure is between 0 and 999.99. A family whose expenditure is 1000 would be included in the class interval 1000-2000.

Inclusive method:

In this method, the overlapping of the class intervals is avoided. Both the lower and upper limits are included in the class interval. This type of classification may be used for a grouped frequency distribution for discrete variable like members in a family, number of workers in a factory etc., where the variable may take only integral values.

It cannot be used with fractional values like age, height, weight etc. In case of continuous variables, the exclusive method should be used. The inclusive method should be used in case of discrete variable.

Open end classes:

A class limit is missing either at the lower end of the first class interval or at the upper end of the last class interval or both are not specified. The necessity of open end classes arises in a number of practical situations, particularly relating to economic and medical data when there are few very high values or few very low values which are far apart from the majority of observations.

Class limit: it represents the end points of a class interval. {Lower class limit & Upper class limit}. A class interval which has neither upper class limit nor lower class limit indicated is called an open class interval e.g. "less than 25", "25 and above"

Cumulative frequency: This is the sum of a frequency of the particular class to the frequencies of the class before it.

Example 1. The following are the marks of 50 students :

48 70 60 47 51 55 59 63 68 63 47 53 72 53 67 62 64 70 57 56 48 51 58 63 65 62 49 64
53 59 63 50 61 67 72 56 64 66 49 52 62 71 58 53 63 69 59 64 73 56.

- (a) Construct a frequency table for the above data.
- (b) Answer the following questions using the table obtained:
 - (i) how many students scored between 51 and 62?
 - (ii) how many students scored above 50?
 - (iii) what is the probability that a student selected at random from the class will score less than 63?

Solution 1 (a) $Range (R) = 73 - 47 = 26$

$$\# \text{ of classes } (k) = \sqrt{n} = \sqrt{50} = 7.07 \approx 7$$

$$\text{Class size } (w) = \frac{R}{k} = \frac{26}{7} = 3.7 \approx 4$$

Frequency table

<i>Mark</i>	<i>frequency</i>
47-50	7
51-54	7
55-58	7
59-62	8
63-66	11
67-70	6
71-74	4
Σ	50

(b) (i) 22 (ii) 43 (iii) 0.58

Example 2. The following data represent the ages (in years) of people living in a housing estate :

18 31 30 6 16 17 18 43 2 8 32 33 9 18 33 19 21 13 13 14 14 6 52 45 61 23 26 15 14 15
14 27 36 19 37 11 12 11 20 12 39 20 40 69 63 29 64 27 15 28.

Present the above data in a frequency table showing the following columns; class interval, class mark (mid-point), frequency and cumulative frequency in that order.

Solution 2 $\text{Range } (R) = 69 - 2 = 67$

$$\# \text{ of classes } (k) = \sqrt{n} = \sqrt{50} = 7.07 \approx 7$$

$$\text{Class size } (w) = \frac{R}{k} = \frac{67}{7} = 9.5 \approx 10$$

<i>class interval</i>	<i>class mark</i>	<i>frequency</i>	<i>cumulative frequency</i>
<i>2-11</i>	<i>6.5</i>	<i>7</i>	<i>7</i>
<i>12-21</i>	<i>16.5</i>	<i>21</i>	<i>28</i>
<i>22-31</i>	<i>26.5</i>	<i>8</i>	<i>36</i>
<i>32-41</i>	<i>36.5</i>	<i>7</i>	<i>43</i>
<i>42-51</i>	<i>46.5</i>	<i>2</i>	<i>45</i>
<i>52-61</i>	<i>56.5</i>	<i>2</i>	<i>47</i>
<i>62-71</i>	<i>66.5</i>	<i>3</i>	<i>50</i>

1.6.2 GRAPHICAL PRESENTATION OF DATA

It is not enough to represent data in a tabular form. The most attractive way of representing data is through charts or graphs.

PIE CHART

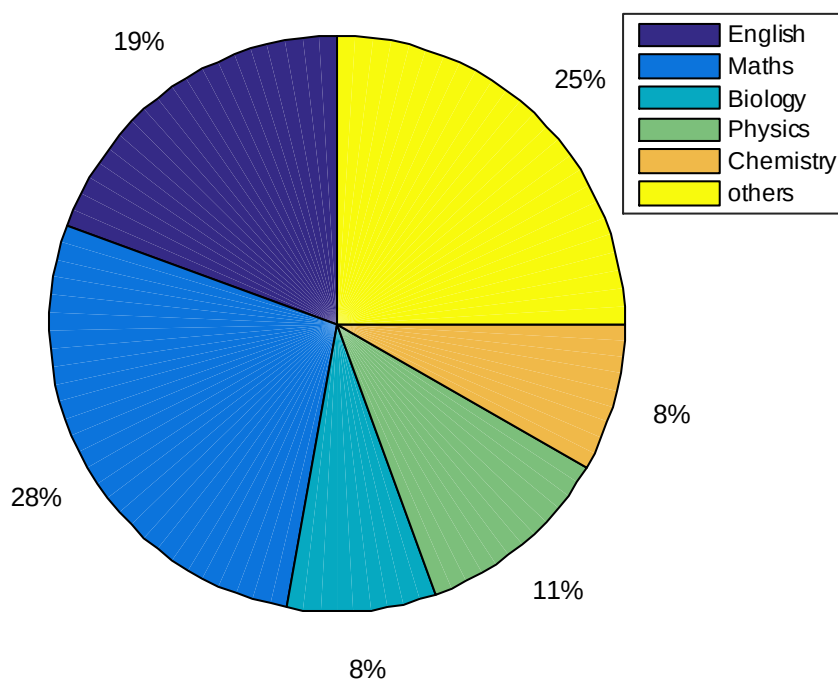
A pie chart is a circular graph in which numerical data are represented by sectors of a circle. The angles of the sectors are proportional to the frequencies of the items they represent

Example. In a school, the lesson periods for each week are given below.

English 7, Maths 10, Biology 3, Physics 4, Chemistry 3, others 9. Draw a pie chart to illustrate this information.

Solution 3 *Total no. of periods in a week = $7+10+3+4+3+9 = 36$*

<i>Subject</i>	<i>N° of Periods</i>	<i>Angle of sector</i>
<i>English</i>	<i>7</i>	$\frac{7}{36} \times 360^\circ = 70^\circ$
<i>Maths</i>	<i>10</i>	$\frac{10}{36} \times 360^\circ = 100^\circ$
<i>Biology</i>	<i>3</i>	$\frac{3}{36} \times 360^\circ = 30^\circ$
<i>Physics</i>	<i>4</i>	$\frac{4}{36} \times 360^\circ = 40^\circ$
<i>Chemistry</i>	<i>3</i>	$\frac{3}{36} \times 360^\circ = 30^\circ$
<i>others</i>	<i>9</i>	$\frac{9}{36} \times 360^\circ = 90^\circ$
Σ	<i>36</i>	360°

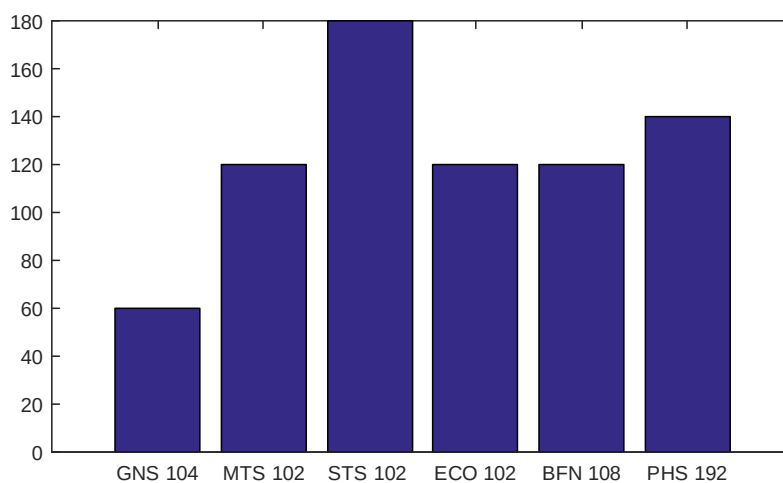


BAR CHART

A bar chart is a statistical graph in which bars (rectangular bars) are drawn such that their lengths or heights are proportional to the quantities or item they represent. Each bar is separated by equal gaps.

Example. The allotment of time in minutes per week for some of the university courses in second semester is :

Courses	Minutes
GNS 104	60
MTS 102	120
STS 102	180
ECO 102	120
BFN 108	120
PHS 192	140



HISTOGRAM

Histograms are similar to bar charts; they are a way to display counts of data. A bar graph charts actual counts against categories; The height of the bar indicates the number of items in that category. A histogram displays the same categorical variables in “bins”.

A bin shows how many data points are within a range (an interval). Normally, you choose the range that best fits your data. There are no set rules about how many bins you can have, but the rule of thumb is 5-20 bins. Any more than 20 bins and your graph will be hard to read. Fewer than 5 bins and your graph will have little (if any) meaning.

What does the height of a bar in a histogram represent?

Unlike a bar chart, the area of a bar in a histogram represents the frequency, not the height. The frequency is calculated by multiplying the width of the bin by the height. The height of a bar in a histogram indicates frequency (counts) only if the bin widths are evenly spaced. For example, if you are plotting magnitudes of earthquakes and your bins are 3-5, 5-7 and 7-9, each bin is spaced two numbers apart and so the height of the bar would equal the frequency. However, histograms don't always have even bins. When a histogram has uneven bins, the height does not equal the frequency.

The table below gives the marks of 80 students on an exam. The data has already been grouped for us into 10 classes. The exam scores are given in whole marks.

Range of marks	Frequency
1–10	2
11–20	2
21–30	4
31–40	6
41–50	7
51–60	8
61–70	15
71–80	22
81–90	10
91–100	4
Total	80

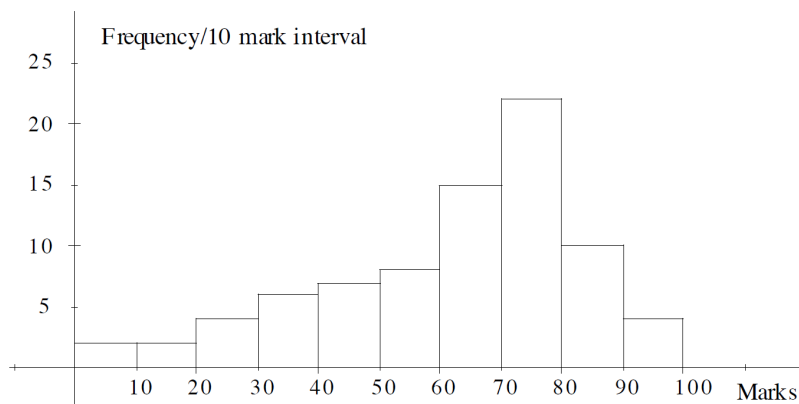
Each of the intervals from 1–10 marks, 11–20 marks and so on is called a class interval. In this example, each class interval is an interval of 10 marks, namely the marks 1 to 10 including both 1 and 10, or 11 to 20 including both 11 and 20, etc. The table tells us that, for example, the class interval 21–30 has a frequency of 4. This means that 4 students scored marks between 21 and 30 inclusive but we don't know their exact marks. A histogram of these data has been drawn below.

Here we have used the right hand endpoint of the class intervals to indicate our horizontal scale. All the class intervals have the same width, 10 marks.

The height of each column represents the frequency per 10 mark interval.

The area of each column represents the number of members in each class interval, or frequency. For the interval 21–30, Area of the rectangle = no. of 10 mark intervals $\times$ frequency/10 mark interval = $1 \times 4 = 4$.

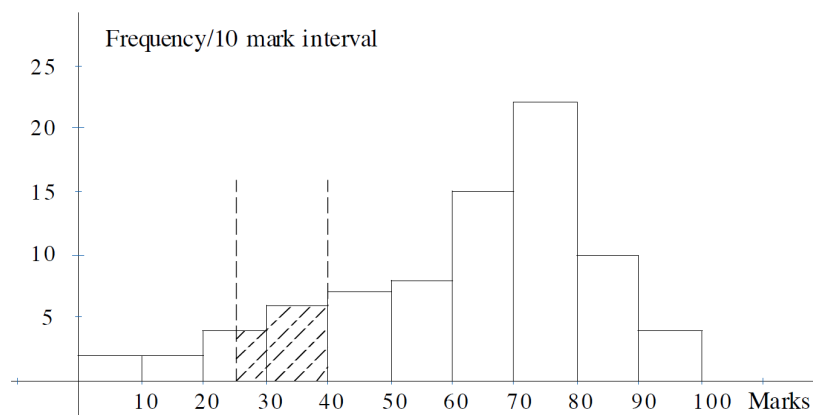
Since each column has the same width, i.e. one, its height is equal to its area. The



total area enclosed represents the total number in the sample.

If we are given a histogram, we can use it to get information about the sample. For example, we can use the histogram in the above figure to estimate the number of people with marks between 26 and 40. We want to find the area of the histogram between the two dotted lines in the next figure. The area is shaded to help you.

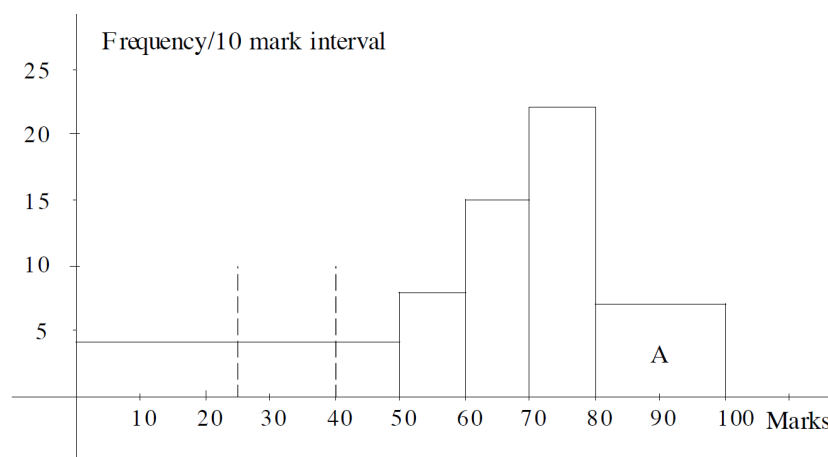
This area is $(\frac{1}{2} \times 4) + (1 \times 6) = 8$. That is, we take half the area of the rectangle 21–30 and add the area of the rectangle 31–40. So we estimate 8 people have marks between 26 and 40.



Now suppose the information is grouped differently.

Range of marks	Frequency	Frequency per 10 marks
1–50	21	4.2
51–60	8	8
61–70	15	15
71–80	22	22
81–100	14	7
Total	80	

Here all marks of 50 and below are grouped in one class interval, and marks above 80 are also grouped together. In drawing this histogram it is extremely important that the area of each column, rather than its height, represents the frequency. The correct units for the vertical axis is again frequency/10 mark interval. The histogram for these data is drawn below.



For example, the number of people who obtained more than 80 marks is the area of rectangle A.

Area of rectangle A = no. of 10 mark intervals $\times$ frequency/10 mark interval = 2×7
= 14.

This histogram can also be used to estimate the number of people with marks between 26 and 40. Again we find the area enclosed by the dotted lines drawn at 25 and 40. This time the estimate is $1.5 \times 4.2 = 6.3$, so our estimate would be 6.

Ogive Graph / Cumulative Frequency Polygon

An ogive (oh-jive), sometimes called a cumulative frequency polygon, is a type of frequency polygon that shows cumulative frequencies. In other words, the cumulative percents are added on the graph from left to right.

An ogive graph plots cumulative (relative) frequency on the y-axis and class boundaries along the x-axis. It's very similar to a histogram, only instead of rectangles, an ogive has a single point marking where the top right of the rectangle would be. It is usually easier to create this kind of graph from a frequency table.

Let's draw an Ogive graph for the set of data given before

Range of marks	Frequency	Relative frequency	Cumulative relative frequency
1–10	2	0.025	0.025
11–20	2	0.025	0.05
21–30	4	0.05	0.1
31–40	6	0.075	0.175
41–50	7	0.0875	0.2625
51–60	8	0.1	0.3625
61–70	15	0.1875	0.55
71–80	22	0.275	0.825
81–90	10	0.125	0.95
91–100	4	0.05	1
Total	80		

